

AI/ML Internship – Day 73 Notes & Tasks

Module 9: Retrievers in RAG – Dense Retrieval, Sparse Retrieval, BM25 & Hybrid Search

Part 1: Recap of Day 72

Yesterday we learned:

-  Similarity Search
-  Cosine Similarity
-  Euclidean Distance
-  Dot Product
-  Top-K Retrieval

Today we will learn:

- What is a Retriever?
 - Types of Retrievers
 - Sparse Retrieval
 - Dense Retrieval
 - BM25
 - Hybrid Search
 - Choosing the Right Retriever
-

Part 2: What is a Retriever?

Definition

A **Retriever** is a component in a RAG system that searches a knowledge base and retrieves the most relevant documents or text chunks for a user's query.

The retrieved information is then passed to the Large Language Model (LLM) to generate the final answer.

Simple Definition

A retriever acts like a **smart librarian**.

Imagine you ask a librarian:

"Can you find the AI book that explains Machine Learning?"

The librarian searches the library and gives you the most relevant books.

Similarly, a retriever searches the vector database and returns the most relevant document chunks.

Part 3: Where Does the Retriever Work?

User Question

|



Generate Embedding

|



Retriever

|



Vector Database

|



Retrieve Relevant Chunks

|



LLM

|



Final Answer

Part 4: Types of Retrievers

There are four commonly used retrievers:

1. Sparse Retriever
2. Dense Retriever
3. BM25 Retriever
4. Hybrid Retriever

Part 5: Sparse Retrieval

Definition

Sparse Retrieval searches documents based on **exact words or keywords**.

It does not understand the meaning of the sentence.

Example

Document:

Python is a programming language.

User Question:

Python programming

The document is retrieved because it contains the exact keywords.

Advantages

-  Fast
-  Easy to implement
-  Works well for keyword searches

Limitations

-  Cannot understand synonyms
 -  Cannot understand semantic meaning
-

Part 6: Dense Retrieval

Definition

Dense Retrieval uses **embeddings** to understand the meaning of text.

Instead of matching words, it matches semantic similarity.

Example

Document:

The internship lasts for three months.

User Question:

How long is the internship?

Although the words are different, the meaning is the same.

The Dense Retriever retrieves the correct chunk.

Advantages

- ✓ Understands semantic meaning
 - ✓ Better accuracy
 - ✓ Excellent for AI chatbots
-

Limitations

- ✗ Requires embedding models
 - ✗ More computationally expensive
-

Part 7: BM25 Retriever

What is BM25?

BM25 (Best Matching 25) is an improved keyword-based retrieval algorithm.

It ranks documents based on:

- Keyword frequency

- Document length
 - Word importance
-

Example

Query:

Machine Learning

BM25 gives higher scores to documents where:

- "Machine Learning" appears more frequently.
 - The document is relevant without being unnecessarily long.
-

Advantages

- ✓ Better than simple keyword search
 - ✓ Widely used in search engines
-

Limitations

- ✗ Still relies on keywords
 - ✗ Cannot fully understand semantic meaning
-

Part 8: Hybrid Search

Definition

Hybrid Search combines:

- Dense Retrieval
- Sparse Retrieval (or BM25)

The results from both methods are combined to improve retrieval quality.

Example

User Question:

Track my order.

Document:

Your package has been shipped.

- Sparse Retrieval may miss the document because the keywords differ.
- Dense Retrieval understands that "track order" and "package shipped" have similar meanings.

Hybrid Search combines both methods to provide more accurate results.

📌 Part 9: Comparison Table

Feature	Sparse	Dense	BM25	Hybrid
Uses Keywords	✓	✗	✓	✓
Uses Embeddings	✗	✓	✗	✓
Understands Meaning	✗	✓	✗	✓
Speed	Fast	Moderate	Fast	Moderate
Accuracy	Medium	High	High	Very High

📌 Part 10: Retrievers in LangChain

Example:

```
retriever = vectorstore.as_retriever(  
    search_kwargs={"k":3}  
)
```

```
docs = retriever.invoke(  
    "What is the internship duration?"  
)
```

Explanation

- `as_retriever()` converts the vector database into a retriever.
- `k=3` retrieves the top three relevant document chunks.

📌 Part 11: Why Retrievers are Important in RAG

Without a retriever:

✗ The LLM would not know which document chunks are relevant.

With a retriever:

✓ Only the most relevant information is provided to the LLM.

This improves:

- Accuracy
- Speed
- Context quality

📌 Part 12: Real-World Applications

Customer Support

Retrieve FAQ documents.

Healthcare

Find treatment guidelines.

Banking

Search loan policies.

Legal

Retrieve contract clauses.

Education

Search textbooks and lecture notes.

📌 Part 13: Advantages

- ✓ Faster document retrieval
 - ✓ Better AI responses
 - ✓ Improves RAG accuracy
 - ✓ Reduces irrelevant context
-

📌 Part 14: Limitations

- ✗ Poor retrievers lead to poor answers.
 - ✗ Large document collections require optimization.
 - ✗ Hybrid search can require more resources.
-

📌 Part 15: Theory Questions

What is a Retriever?

A retriever searches a knowledge base and retrieves the most relevant document chunks for a user's query.

What is Dense Retrieval?

Dense Retrieval uses embeddings to retrieve documents based on semantic meaning.

What is Sparse Retrieval?

Sparse Retrieval retrieves documents using exact keyword matching.

What is BM25?

BM25 is an advanced keyword-ranking algorithm that considers keyword frequency and document length to rank search results.

What is Hybrid Search?

Hybrid Search combines keyword-based retrieval and embedding-based retrieval to improve search accuracy.

Part 16: Interview Questions

What is the role of a retriever in a RAG system?

The retriever identifies and returns the most relevant document chunks, which are then provided to the LLM as context.

Why is Dense Retrieval better than Keyword Search?

Dense Retrieval understands semantic meaning, allowing it to retrieve relevant information even when different words are used.

What is the difference between BM25 and Dense Retrieval?

BM25 ranks documents based on keyword frequency and relevance, while Dense Retrieval compares semantic embeddings.

Why do companies use Hybrid Search?

Hybrid Search combines the strengths of keyword search and semantic search, resulting in more accurate and reliable retrieval.

Which retriever is commonly used in modern RAG systems?

Dense Retrieval and Hybrid Search are the most commonly used approaches in modern enterprise RAG applications.

Tasks for Day 73

Theory

1. Define a Retriever.
2. Differentiate Sparse Retrieval and Dense Retrieval.
3. Explain BM25.
4. What is Hybrid Search?
5. Why are retrievers important in RAG?

Practical Task 1

Create a ChromaDB collection and convert it into a retriever using LangChain.

Practical Task 2

Retrieve the **Top 3** document chunks for five different user questions.

Practical Task 3

Compare the results of:

- Keyword Search
- Dense Retrieval

Write down which performs better and explain why.

Practical Task 4

Research BM25 and list two real-world search engines or applications that use keyword-ranking techniques.

Practical Task 5

Create a comparison table for:

- Sparse Retrieval
- Dense Retrieval
- BM25
- Hybrid Search

Include their advantages, disadvantages, and use cases.